Random Lattice Graphs

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Abstract

We define an integer lattice of radius r, build graphs constructed on this bounded lattice space, and explore randomized graph construction on the space.

1 The Bounded Integer Lattice

The 2-dimensional integer lattice is define as the set $\mathrm{Z}^2 = \mathrm{Z} \times \mathrm{Z}$. We define the finite and bounded subset of this lattice with radius r as:

$$L_{\mathrm{Z}^2}(r) = Z^2 \cap [-r, r]^2 \tag{1}$$

Equivalently, it is the collection of points $(x, y) \in \mathrm{Z}^2$ for which $x, y \in [-r, r]$. For the rest of the paper, assume that we are always working with the 2-dimensional integer lattice, so we will continue with the notation of L, for the complete 2-dimensional lattice, and $L(r)$ for the sub-lattice of radius r as defined above.

It is clear that the standard notion of distance in between two points in $L(r)$, just as in L, is that of the Manhattan distance defined by:

$$d(x, y) = |x_1 - y_1| + |x_2 - y_2|$$

This is a path metric and is used to define the norm of any one point from the origin, denoted by $||(x, y)||_{L(r)}$. While we define the distance this way, the geometry of our sub-lattice is that of a discrete box bounded by the L^∞ norm:

$$L(r) = \{(x, y) \in \mathrm{Z}^2 : ||(x, y)||_{\mathrm{L}^\infty} \leq r\}$$

Clearly, $L(r)$ is finite and bounded by L^1 (Manhattan) metric. As for the cardinality, we have $|L(r)| = (2r + 1)^2$.

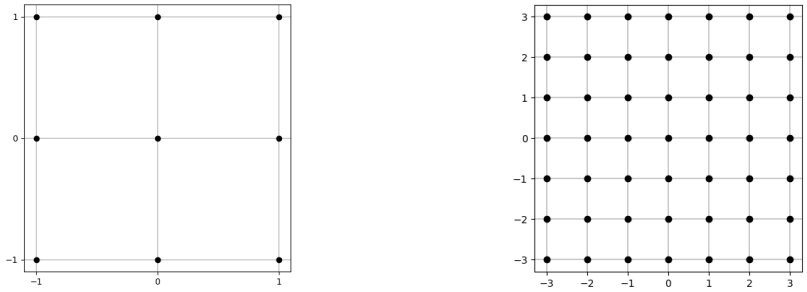


Figure 1: The $L(1)$ and $L(3)$ lattices.

2 Graphs Embedded in $L(r)$

Constructing a graph over/embedded in $L(r)$ simply means that all vertices of the graph are points in $L(r)$, and the "length" of edges connecting any two vertices follow the minimal path with respect to the L^1 metric. That is, a graph $G = (V, E)$ is embedded in $L(r)$ if:

- $V \subseteq L(r)$
- $l(e) = d_{\mathrm{L}^1}(u, v)$ for all edges $e = (u, v) \in E$

Though, the length of edges and their exact path can be left abstract, unless absolutely necessary, similar to unweighted graphs, where the design of any edge is completely arbitrary and left only to fill the abstract notion of connectedness.

2.1 Isomorphisms and Embeddings

Even if there exists some graph G' that is not necessarily embedded in $L(r)$, there are particular conditions that enable G' to be represented in $L(r)$. Assume $G' = (V', E')$ is some unweighted graph. In this case, edge length is negligible, and as long as the following holds:

$$|L(r)| \geq |V'|,$$

there exists a graph $G \in L(r)$ that is isomorphic to G'. Because the above condition is satisfied, there must exist an injective mapping $\phi : V' \to L(r)$, giving this isomorphism. As for the edges, if $(u', v') \in G'$, we simply reconnect the images of these vertices in G:

$$(u, v) \in G' \implies (\phi(u), \phi(v)) \in G \tag{2}$$

From the previous section, we know that $|L(r)| = (2r + 1)^2$. Thus, for any graph G', not necessarily embedded in $L(r)$, with n vertices, we require at least $L(\lceil(\sqrt{n} - 1)/2\rceil)$ to embed it into some finite, bounded lattice.

Theorem. For all unweighted $G' = (V', E')$ with $|V'| = n$, there exists a graph $G \in L(r)$ with $r \geq \lceil(\sqrt{n} - 1)/2\rceil$ such that $G' \cong G$.

Now, what if we want to embed some *weighted* graph G' into some lattice $L(r)$? From the above, we know how minimal r can be, but if $\max_{e \in E'} l(e) > 4r$, we cannot perform the embedding. Thus, there are many more constraints we need to cover for weighted graphs in order to draw an existence of an isomorphic counterpart $G \in L(r)$.

3 Random Graphs Embedded in $L(r)$

We begin the section by building a notion of randomized graph embedded in $L(r)$. We must choose a random process that constructs both V and E. Let us construct $G = (V, E)$ using the following: $u \in L(r)$ is apart of V with probability p_v, and an edge connects any two points/vertices u, v with probability p_e. From this simple random construction, the expected number of vertices n is:

$$\mathbb{E}[n] = p_v|L(r)|$$

As for edges, let us first look at the expected degree of any particular vertex u. For any vertex $u \in G$, there are $p_v|L(r)| - 1$ choices for connection by the above. As an edge exists between any two vertices with probability p_e, we expect the degree of a given vertex to be:

$$\mathbb{E}[\deg(u)] = p_e(p_v|L(r)| - 1)$$

Similar to the way we derive the expected number of vertices, we can consider the set of all possible edges, and multiply it by the probability p_e to achieve the expected number of edges for a random graph embedded in $L(r)$ generated by the random process described initially. For any graph G with n vertices, the number of possible edges without duplication is $n(n+1)/2$. Let n be our expected number of vertices as previously described, then the expected number of edges must be:

$$\mathbb{E}\left[|E|\right] = p_e \left(\frac{n(n+1)}{2} \right) \tag{3}$$

$$= p_e \left(\frac{(p_v|L(r)|)(p_v|L(r)|+1)}{2} \right) \tag{4}$$

$$= p_e \left(\frac{(p_v|L(r)|^2 + p_v|L(r)|}{2} \right) \tag{5}$$

We may also simply write it in terms of expected values:

$$\mathbb{E}[|E|] = p_e \left(\frac{\mathbb{E}[|V|]^2 + \mathbb{E}[|V|]}{2} \right)$$